

Linear Algebra

1. (August 2017 Problem 2) Let K be a field, and let A be an $n \times n$ matrix over K . Suppose that $f(x) \in K[x]$ is an irreducible polynomial such that $f(A) = 0$. Show that $\deg(f) \mid n$.

1. Solution Let $m(x) \in K[x]$ be the minimal polynomial of A . Since $f(A) = 0$, $m(x)$ must divide $f(x)$, but $f(x)$ is irreducible so $m(x) = f(x)$. Let $\chi(x)$ be the characteristic polynomial of A , which has degree n . Since $\chi(x)$ and $f(x)$ have the same roots and $f(x)$ is irreducible, it must be the case that $\chi(x) = f(x)^\ell$ for some ℓ . Therefore, $\ell \cdot \deg(f) = \deg(\chi) = n$, so $\deg(f) \mid n$.

2. Give two examples of matrices in Jordan form with minimal polynomial $x(x-1)^2(x-2)$ and characteristic polynomial $x^2(x-1)^4(x-2)$.

2. Solution We are looking for 7×7 matrices that have eigenvalues 0, 1, and 2 with maximal Jordan blocks of size 1, 2, and 1 respectively. Here are two such matrices:

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}$$

3. Let V be a finite-dimensional vector space over a field. Let $A : V \rightarrow V$ be an endomorphism, and suppose $W \subseteq V$ is an invariant subspace (so that $A(W) \subseteq W$). Let $\bar{A} : V/W \rightarrow V/W$ be defined by $\bar{A}(v + W) = Av + W$ for all $v \in V$.

(a) Show that $\bar{A}$ is well-defined.

(b) Show that if $\bar{A}$ is not invertible, then A is not invertible.

3. Solution

(a) We need to check that if v and v' are two representatives of the same element in V/W , then $\bar{A}(v + W) = \bar{A}(v' + W)$. Since v and v' represent the same coset, we have $v' = v + w$ for some $w \in W$. Then by definition, we have

$$\bar{A}(v' + W) = Av' + W = A(v + w) + W = Av + Aw + W = Av + W = \bar{A}(v + W),$$

where the second-to-last equality follows since $Aw \in W$.

(b) We prove the contrapositive. Suppose A is invertible with inverse B . We will show that $\bar{A}$ is invertible by constructing an inverse. First, we show that $B(W) \subseteq W$. Choose a basis $w_1, \dots, w_r$ for W , and let $w'_i = Aw_i$. Since A is invertible, the w'_i are linearly

independent, and since $AW \subseteq W$, $w'_i \in W$. Thus, the w'_i form a basis for W . We also know that $Bw'_i = w_i \in W$, and so since every element of W is a linear combination of the w'_i , $BW \subseteq W$. (If you can find an elegant basis-free way to write this, I'd love to know it.)

Define $\overline{B} : V/W \rightarrow V/W$ by $\overline{B}(v + W) = Bv + W$. As in the previous part, this is well-defined and is indeed an inverse for $\overline{A}$ since for any $v + W \in V/W$ we have $\overline{BA}(v + W) = \overline{B}(Av + W) = BAv + W = v + W$ so $\overline{BA} = Id$, and similarly $\overline{AB} = Id$.

4. (January 2020 Problem 4) Let V be a finite-dimensional vector space over a field. Let $A : V \rightarrow V$ be a linear transformation.

- (a) Show that there exists $B : V \rightarrow V$ such that $ABA = A$.
- (b) Show that there exists $C : V \rightarrow V$ such that $ACA = A$ and $CAC = C$.

4. Solution

- (a) Let $v_1, \dots, v_r$ be a basis for $\text{im } A$, and let $\tilde{v}_1, \dots, \tilde{v}_r$ be any preimages of $v_1, \dots, v_r$. Extend $v_1, \dots, v_r$ to a basis for V arbitrarily, and define B by $Bv_i = \tilde{v}_i$ for $1 \leq i \leq r$ and $Bv_i = 0$ otherwise. Then AB is the identity on $\text{im } A$ and 0 otherwise. Hence, $ABA = A$.
- (b) Let $C = B$ as constructed above. By the same token, CA is the identity on $\text{im } C$, so $CAC = C$.

5. (January 2015 Problem 4) Let F be a field, and let V be a finite-dimensional vector space over F that has dimension $n \geq 1$. Let q be a nonzero element of F such that $q^i \neq 1$ for $1 \leq i \leq n$. Let $X : V \rightarrow V$ and $Y : V \rightarrow V$ be two F -linear transformations such that $XY = qYX$. Show that XY is nilpotent.

5. Solution

Suppose $\lambda \neq 0$ is an eigenvalue of XY . Then $XYv = \lambda v$ for some nonzero $v \in V$, and since $XY = qYX$, we also have that $XY(Yv) = qYXYv = qY(XYv) = q\lambda Yv$. So, $q\lambda$ is also an eigenvalue for XY with eigenvector Yv . (Note that $Yv \neq 0$ since $XYv \neq 0$.) Iterating this gives that $q^i\lambda$ is an eigenvalue of XY for each $0 \leq i \leq n$. Since $q^i \neq 1$ for $1 \leq i \leq n$, this means that XY has at least $n + 1$ distinct eigenvalues, which is impossible since V is n -dimensional. Therefore, $\lambda = 0$ is the only eigenvalue of XY , and hence XY is nilpotent.

6. Let V, W be two finite-dimensional vector spaces over $\mathbb{C}$. Let $A : V \rightarrow V$ and $B : W \rightarrow W$ be linear transformations. List the eigenvalues of $A \otimes B : V \otimes W \rightarrow V \otimes W$ in terms of the eigenvalues of A and B .

6. Solution We can suppose that A and B are in Jordan canonical form. Then A and B are upper triangular (with respect to particular bases on V and W) with diagonal elements $a_{11}, \dots, a_{nn}$ and $b_{11}, \dots, b_{mm}$ where $\dim(V) = n$ and $\dim(W) = m$. Then $V \otimes W$ has a basis given by the tensor product of the chosen bases for V and W , and $A \otimes B$ is an upper triangular matrix with diagonal elements $a_{ii}b_{jj}$ for $1 \leq i \leq n$ and $1 \leq j \leq m$. So, these are

the eigenvalues of $A \otimes B$.

7. (January 1994 Problem 4) Let X be a subspace of $M_n(\mathbb{C})$, the $\mathbb{C}$ -vector space of all $n \times n$ complex matrices. Assume that every nonzero matrix in X is invertible. Prove that $\dim_{\mathbb{C}}(X) \leq 1$.

7. Solution Let A and B be arbitrary matrices in X . Since they are both invertible, consider the matrix AB^{-1} . Since we are over $\mathbb{C}$ this matrix must have an eigenvalue λ . So, $AB^{-1} - \lambda \text{Id}$ is not invertible, which means that $\det(AB^{-1} - \lambda \text{Id}) = 0$. Then we have

$$\det(A - \lambda B) = \det(AB^{-1} - \lambda \text{Id}) \det(B) = 0$$

which implies that $A - \lambda B = 0$ since it is an element of X that is not invertible. Thus, $A = \lambda B$. Since all of the matrices in X are scalar multiples of each other, we have that $\dim_{\mathbb{C}}(X) \leq 1$.

8. (January 2017 Problem 2) Let $n > 0$ be an integer. Let F be a field of characteristic zero, V a vector space over F of dimension n , and $T : V \rightarrow V$ an invertible F -linear map such that $T^{-1} = T$. Denote by W the vector space of linear transformations from V to V that commute with T . Find a formula for $\dim(W)$ in terms of n and the trace of T .

8. Solution Since $T^{-1} = T$, we see that $T^2 - \text{Id}_n = 0$, so T satisfies the polynomial $x^2 - 1 \in F[x]$. Therefore, the minimal polynomial of T must divide $x^2 - 1 = (x - 1)(x + 1)$. So the only possible eigenvalues of T are 1 and -1 . Since the minimal polynomial of T has distinct roots, we can choose a basis so that T is a diagonal matrix with only 1's and -1 's on the diagonal. Furthermore, we can rearrange the basis so that T has the block matrix form

$$T = \begin{pmatrix} \text{Id}_a & 0 \\ 0 & -\text{Id}_b \end{pmatrix}$$

where $a + b = n$. So the trace of T is $\text{tr}(T) = a - b$.

We want to understand matrices that commute with T . Given an $n \times n$ matrix S , we consider it as a block matrix of the form $S = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ where A has size $a \times a$ and D has size $b \times b$. Then multiplying by T on either side gives

$$TS = \begin{pmatrix} A & B \\ -C & -D \end{pmatrix} \quad \text{and} \quad ST = \begin{pmatrix} A & -B \\ C & -D \end{pmatrix}.$$

Therefore, $S \in W$ if and only if $B = -B = 0$ and $C = -C = 0$. Since there are no constraints on A and D , we see that $\dim(W) = a^2 + b^2 = \frac{1}{2}((a + b)^2 + (a - b)^2) = \frac{1}{2}(n^2 + \text{tr}(T)^2)$.

9. (January 2019 Problem 5) Let F be a field and n be a positive integer. Fix an $n \times n$ matrix S over F this is invertible and symmetric. Writing A^t for the transpose of a matrix, we let

$$V := \{n \times n \text{ matrices } A \text{ over } F \mid A^t = SAS^{-1}\}.$$

Note that V is a vector space (you do not need to prove this). Find the dimension of V in terms of n .

9. Solution Since S is an arbitrary invertible symmetric matrix, let's first think about V when $S = \text{Id}_n$. In this case, V is the space of matrices that are equal to transpose, i.e. those that are symmetric. Thus, $\dim(V) = n^2 + (n-1)^2 + \cdots + 1 = \frac{1}{2}(n^2 + n)$. Now the question is, why is this the dimension of V for any invertible symmetric S ?

Observe that $A^t = SAS^{-1}$ if and only if $A^t S = SA$, and since $S^t = S$, this is equivalent to $(SA)^t = SA$. So, we want to find the dimension of the space of matrices A such that SA is symmetric. Consider the map $M : M_n(F) \rightarrow M_n(F)$ defined by $A \mapsto SA$. Since S is invertible, this map is an isomorphism. Therefore, $\dim(V)$ is the dimension of the space of symmetric matrices, which is exactly $\frac{1}{2}(n^2 + n)$ as before.